

12.421/12.621 Lecture 9

Volume Scattering

Topics covered

- Fundamentals of volume scattering
- Heuristic scattering from vegetation
- Heuristic scattering from snow

Basics of volume scattering from heterogeneous media

- Distinction between surface and volume scattering
 - Surface scattering
 - Reflectivity a function of
 1. Dielectric constant of surface
 2. Tilting of surface relative to plane of incidence (i.e., long-wavelength ($\gg \lambda$) surface height variations)
 - Directionality of scattering, only function of surface roughness
 - Volume scattering
 - Reflectivity function of
 1. Dielectric discontinuities within the volume
 2. Density of dielectric heterogeneities
 - Directionality of scattering function of
 1. Roughness of boundary surface
 2. Average dielectric constant of media
 3. Size of inclusions
- How we think about and model volume scattering depends primarily on
 - Size distribution of scatterers (relative to λ)
 - Large scatterers will be treated differently from small scatterers
 - Orientation of scatterers
 - Shape distribution of scatterers
 - dielectric contrast between scatterers and background material

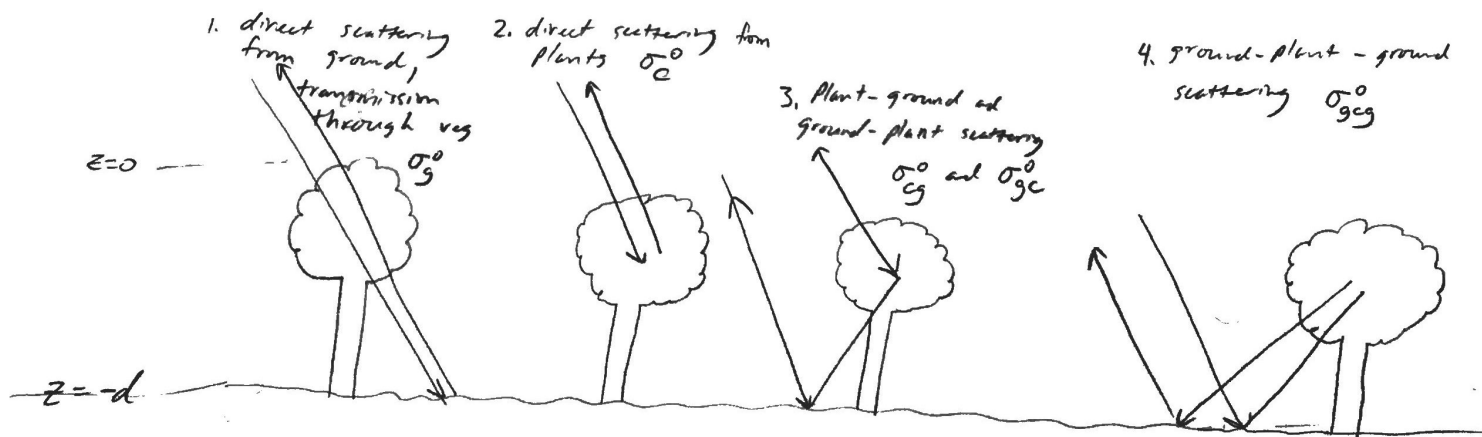
Basics of volume scattering (cont.)

• The key questions

- What approximations are applicable? E.g.,
 - Large, randomly oriented cylinders (often good for vegetation)
 - Small spherical scatterers (e.g., snow, atmosphere, etc.)
- Can we assume the volume layer acts as a single scatterer?
- Are losses dominated by absorption or scattering?
- Azimuthal symmetry?

Heuristic model for vegetation

- "Single-scattering model" which means only one (or zero) bounces within the volume & can have up to 3 total bounces when counting surface scattering from the ground
- Expected scattering types and paths



- Key point: Top of vegetation ($z=0$) is diffuse $\Rightarrow$ no scattering; important simplification that will not be valid for all volume scattering (e.g., snow is a volume scatterer w/ surface scattering component)

Heuristic vegetation scattering model (cont.)

- To account for transmission through canopy, define p-polarized extinction coefficient (a.k.a., power attenuation factor)

$$K_e^p = K_a^p + K_s^p \rightarrow \text{related to penetration depth} \rightarrow \sigma_p = \frac{1}{K_e}$$

$$K_a^p \rightarrow \text{absorption losses} \rightarrow K_a^p = -2K_0 \operatorname{Im}\{\sqrt{\epsilon}\} = (\text{cf. } \alpha, \text{ attenuation const, from previous lectures})$$

$$K_s^p \rightarrow \text{scattering losses}$$

In general, K_e^p, K_a^p, K_s^p vary in space and time; particularly w/ z

Hereafter, assume K_e^p, K_a^p, K_s^p are uniform

- Now define transmissivity of canopy

$$T_p = e^{-K_e^p d / \cos \theta_i}$$

- Let us consider each scatterer defined in the previous cartoon

1. Direct scattering from the ground.

Recalling the previous lecture on surface scattering, we can expect a ground scattered component in the absence of vegetation to be $\sigma_{pg}^0(\theta_i)$. With vegetation, we expect some attenuation due to absorption and scattering in the canopy

$$\Rightarrow \sigma_{pg}^0 = T_p T_g \sigma_{pg}^0 \quad p, g = h \text{ or } v$$

2. Direct volume contribution

- There are many ways to model this component
- By far the simplest is to treat as a "cloud"

$$\hat{\sigma}_{vpg}^{\text{back}} = N_v \sigma_{pg}^{\text{back}}$$

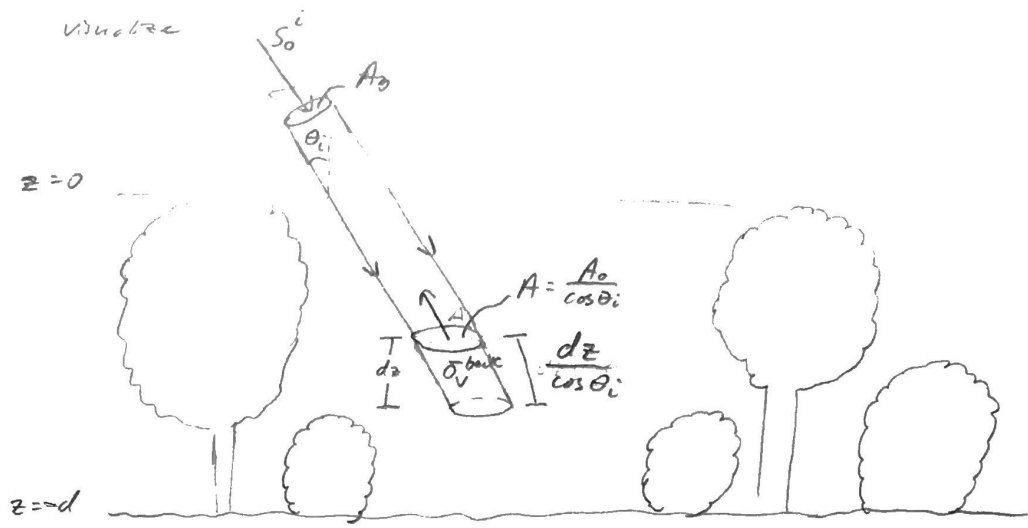
$$\hat{\sigma}_{vpg}^{\text{back}} \equiv \text{volume backscattering coeff. (m}^{-1}\text{)}$$

$$\sigma_{pg}^{\text{back}} \equiv \text{back scattering coeff of a single particle (m}^2\text{)}$$

$$N_v \equiv \text{number of "particles" per unit volume (m}^{-3}\text{)}$$

Heuristic vegetation scattering model (cont.)

2. Direct volume contribution (cont.)



(q-polarized)

- Power density (recall Poynting vector) across area A of a differential volume element within the canopy at depth $z < 0$

$$S_f^i(z) = S_0^i \cos \theta_i e^{-\chi_q^i z / \cos \theta_i} \quad \left[\frac{W}{m^2} \right]$$

$S_0^i \equiv$ incident power density in free space

- Backscattering cross section of differential volume

$$d\sigma_{pq}^{back} = \underbrace{\sigma_{pq}^{back}}_{\text{backscattering cross section / unit volume}} \underbrace{\frac{A dz}{\cos \theta_i}}_{\text{volume of differential element}} \quad [m^2]$$

- p-polarized power reradiated by differential volume, in the backscatter direction, after propagation from depth $z < 0$ to top of canopy ($z=0$)

$$\begin{aligned} dP_p^{rer} &= \int_0^i S_f^i(z) d\sigma_{pq}^{back} e^{-\chi_p^i z / \cos \theta_i} \\ &= S_0^i A \sigma_{pq}^{back} \int_0^i e^{-z(\chi_q^i + \chi_p^i) / \cos \theta_i} dz \end{aligned}$$

Heuristic vegetation scattering model (cont.)

2. Direct volume contribution (cont.)

- Integrating dP_p^{ret} over the canopy depth

$$\begin{aligned}
 P_p^{ret} &= S_0^i A \int_{-d}^0 \hat{\sigma}_{vpq}^{back} e^{-(K_e^p + K_e^s)z/\cos\theta_i} dz \\
 &= \left\{ \frac{S_0^i A \cos\theta_i \hat{\sigma}_{vpq}^{back}}{K_e^p + K_e^s} \right\} \left[1 - e^{-(K_e^p + K_e^s)d/\cos\theta_i} \right] \\
 &= \left\{ \downarrow \right\} [1 - \tau_p \tau_s]
 \end{aligned}$$

- Defining the normalized radar cross section from the canopy as

$$\sigma_{cpq}^0 = \frac{4\pi r^2}{A} \frac{S_p^s}{S_0^i} \quad r \equiv \text{distance to antenna}$$

where

$$S_p^s = \frac{P_p^{ret}}{4\pi r^2} \quad \text{is the power density at the receiving antenna}$$

we have

$$\begin{aligned}
 \sigma_{cpq}^0 &= \frac{\hat{\sigma}_{vpq}^{back} \cos\theta_i}{K_e^p + K_e^s} (1 - \tau_p \tau_s) \\
 &= \frac{\hat{\sigma}_{vpq}^{back} \cos\theta_i}{K_e^p + K_e^s} (1 - e^{-(K_e^p + K_e^s)d \sec\theta_i})
 \end{aligned}$$

where

$\hat{\sigma}_{vpq}^{back}$, K_e^p and K_e^s are all functions of the density and dielectric properties of the leaves and stems that make up the canopy

Henrici-Otto vegetation scattering model (cont.)

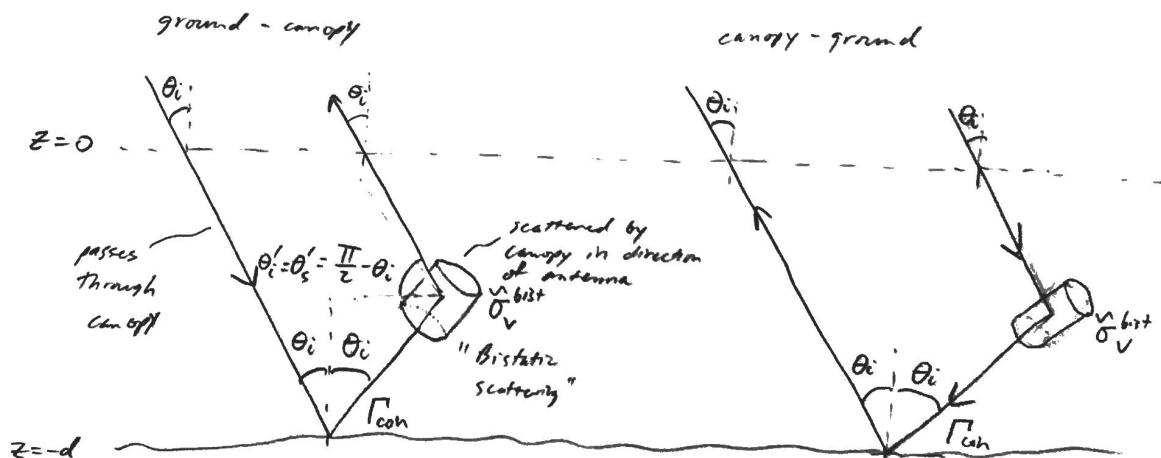
3. Canopy - ground contributions

- Here we have reflection from ground then scattering from canopy and vice versa
- Assume the ground is quasi-specular reflector meaning that only consider reflection in the specular direction but account for diffuse scattering due to surface roughness in the reflectivity
- Assume a very simple model

$$\Gamma_{coh}^p = \Gamma^p(\theta_i) e^{-4K_0^2 h^2 \cos^2 \theta_i}$$

$\Gamma^p(\theta_i) \equiv |p_p|^2$ where $p_p \equiv$ Fresnel reflection coeff
 $h \equiv$ RMS height of surface roughness

Sketch



Normalized radar cross sections

ground - canopy:

$$\sigma_{gc}^0(\theta_i) = \tilde{\sigma}_{V_{gs}}^{bist}(\theta_i) d \Gamma_{coh}^g \gamma_g \gamma_p$$

canopy - ground

$$\sigma_{cg}^0(\theta_i) = \tilde{\sigma}_{V_{gs}}^{bist}(\theta_i') d \Gamma_{coh}^p \gamma_p \gamma_g$$

Heuristic vegetation scattering model (cont.)

3. Canopy-ground (cont.)

- The total canopy-ground contribution

$$\begin{aligned}\sigma_{cgt}^0 &= \sigma_{gc}^0 + \sigma_{cg}^0 \\ &= \hat{\sigma}_{vp}^{bit} d (\Gamma_{coh}^p + \Gamma_{coh}^g) \gamma_p \gamma_g\end{aligned}$$

- In the case of co-polarized scattering ($p=g$), we have:

$$\sigma_{cgt,pp}^0 = 2 \hat{\sigma}_{vp}^{bit} d \Gamma_{coh}^p \gamma_p^2$$

- However, $\uparrow$ not quite correct because we added the powers (for simplicity) rather than taking the power of the total E field. In other words, we undertook incoherent addition of fields. Since ground-canopy and canopy-ground follow same path in reverse, should expect equivalent phase $\Rightarrow$ E-fields add coherently. Can correct by simply multiplying by factor of two

$$\Rightarrow \boxed{\sigma_{cgt,pp}^0 = 4 \hat{\sigma}_{vp}^{bit} d \Gamma_{coh}^p \gamma_p^2 \quad \text{for } p = \text{horv}}$$

4. Ground-canopy-ground contribution

- This is very simple: 2 way transmission, 2 bounces off the ground, and backscatter from the canopy $\sigma_{cg}^0(\theta_i)$

$$\begin{aligned}\Rightarrow \sigma_{gcg,pg}^0 &= \gamma_p \gamma_g \Gamma_{coh}^p \Gamma_{coh}^g \sigma_{cg}^0 \\ &= \frac{\hat{\sigma}_{vp}^{back} \cos \theta_i}{\chi_e^p + \chi_e^g} \Gamma_{coh}^p \Gamma_{coh}^g (\gamma_p \gamma_g - \gamma_p^2 \gamma_g^2)\end{aligned}$$

Heuristic vegetation scattering model (cont.)

★ Putting it all together, the total normalized radar cross section is

$$\begin{aligned}\sigma_{pp}^0 &= \sigma_{sp}^0 + \sigma_{cp}^0 + \sigma_{cgt,pp}^0 + \sigma_{sg,pp}^0 \\ &= \gamma_p \gamma_g \sigma_{sp}^0(\theta_i) + \frac{\hat{\sigma}_{V,pp}^{back} \cos \theta_i}{\chi_e^p + \chi_e^g} (1 - \gamma_p \gamma_g) (1 + \Gamma_{coh}^p \Gamma_{coh}^g \gamma_p \gamma_g) \\ &\quad + C \hat{\sigma}_{V,pp}^{bist} d (\Gamma_{coh}^p + \Gamma_{coh}^g) \gamma_p \gamma_g\end{aligned}$$

where

$$C = \begin{cases} 1 & p \neq g \\ 2 & p = g \end{cases} \quad \begin{array}{l} \text{(cross-pol, incoherent addition)} \\ \text{(co-pol, coherent addition)} \end{array}$$

In general, the terms σ_{sp}^0 , $\hat{\sigma}_{V,pp}^{back}$, $\hat{\sigma}_{V,pp}^{bist}$, $\chi_e^p (\Rightarrow \gamma_p)$, $\chi_e^g (\Rightarrow \gamma_g)$, Γ_{coh}^p , Γ_{coh}^g , and d are not known a priori.

- Simplifications

- Canopy of negligible thickness:

$$\lim_{d \rightarrow 0} \sigma_{pp}^0 = \sigma_{sp}^0 \quad (\text{pure surface scattering from the ground})$$

- Very thick and/or dense canopy

$$\begin{aligned}\gamma_p \approx \gamma_g &\approx 0 \\ \Rightarrow \sigma_{pp}^0 &\approx \frac{\hat{\sigma}_{V,pp}^{back} \cos \theta_i}{\chi_e^p + \chi_e^g}\end{aligned}$$

- Isotropic scatterers

$$\begin{aligned}\hat{\sigma}_V^{back} = \hat{\sigma}_V^{bist} &= \chi_s = \chi_e + 2K_0 \operatorname{Im}\{\sqrt{\epsilon}\} \\ \text{and } \gamma_p &= \gamma_g\end{aligned} \quad \left(\begin{array}{l} \text{note } \chi_e = \chi_s + \chi_a \\ \chi_a = -2K_0 \operatorname{Im}\{\sqrt{\epsilon}\} \end{array} \right)$$

$$\Rightarrow \sigma_{pp}^0 = \gamma^2 \sigma_{sp}^0(\theta_i) + \frac{\chi_s}{\chi_e} \frac{\cos \theta_i}{2} (1 - \gamma^2) (1 + \Gamma_{coh}^2 \gamma^2) + 4\chi_s d \Gamma_{coh}^2 \gamma^2$$

Heuristic vegetation scattering model (cont.)

- Rayleigh scatterers (spheres w/ radius $\ll \lambda$)

$$\hat{\sigma}_V^{\text{back}} = \hat{\sigma}_V^{\text{fwd}} = \frac{3}{2} K_S$$

$$\Rightarrow \sigma_{pp}^0(\theta_i) = \gamma^2 \sigma_{spp}^0 + \frac{3}{4} \frac{K_S}{K_e} \cos \theta_i (1 - \gamma^2) (1 + \Gamma_{\text{coh}}^2 \gamma^2) + 3c K_S d \Gamma_{\text{coh}} \gamma^2$$

- General considerations for models of $\hat{\sigma}_V^{\text{back}}$ and $\hat{\sigma}_V^{\text{fwd}}$
 - density of scatterers
 - size of scatterers relative to EM wave length
 - if size $\geq \lambda$, shape and orientation of scatterers

Heuristic model of scattering from a snow layer of depth d

- The volume scattering will be identical to the vegetation model, but now we must account for reflection and transmission at the air-snow interface
- Straightforward extension of previous model

$$\sigma_{pp}^0(\theta_i) = T_p^p T_g^g (\sigma_{pg}^0 / v_{\text{veg}}) + \sigma_{aspg}^0$$

T_p & T_g are transmissivity of air-snow interface

$\sigma_{pg}^0 / v_{\text{veg}}$ is the model derived previously for vegetation scattering

σ_{aspg}^0 is surface scattering from air-snow interface

Summary

- Volume scattering involves scattering from dielectric discontinuities within a volume
- Reflectivity of a volume is function of
 - Dielectric discontinuities within the volume
 - Density of dielectric heterogeneities
- Directionality of scattering
 - Roughness of boundary
 - Average dielectric const of media
 - Size of inclusions
- Modeling of volume scattering depends on
 - size distribution of scatterers (relative to λ)
 - orientation of scatterers
 - shape of scatterers
 - dielectric contrast
- Multiple scattering contributions from volume
 - E.g., vegetation
 - direct ground w/ 2-way transmission through volume
 - direct canopy contribution
 - ground-canopy and canopy-ground
 - ground-canopy-ground
- For volumes w/ nondiffuse surface, must consider transmission through and scattering from interface